MoodArt: Grounding Artistic Style Transfer in Facial Mood Recognition

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Abstract

Style transfer systems apply artistic transformations to photographs without any connection between the image's content and the style selected, the same filter applies equally to a joyful portrait and a tense street photograph because style selection is disconnected from image semantics. We present MoodArt, a system that closes this gap through affect recognition: a mood classifier analyzes the uploaded photograph, and the detected mood determines which artistic style is applied, grounding the style selection in the emotional content of the image rather than in an arbitrary choice.

1 Introduction

Facial emotion recognition (FER) has long been studied as a standalone classification task, yet its potential as a semantic bridge between visual content and creative output remains largely unexplored. We present MoodArt, a system that uses mood classification as an intelligent intermediary: the emotional signal detected in an uploaded photograph determines which artistic style is applied to it, grounding style transfer in the content of the image rather than in an arbitrary user selection.

The core technical challenge is reliable mood classification from photographs. Facial emotion recognition on FER2013 is a well-studied benchmark, but its seven emotion classes do not map cleanly onto artistic styles. We collapse these into five mood classes (happy, calm, melancholic, energetic, dramatic) and address the resulting class imbalance through balanced undersampling and weighted loss.

We make the following contributions:

1. A relabeling of five FER2013 emotion classes as mood descriptors to align with the artistic style vocabulary of the application.
2. A comparative evaluation of three classifier families (rule-based, classical ML with Optuna hyperparameter search, and fine-tuned VGG16) on the mood classification task.
3. A comparative evaluation of three style transfer methods—Color LUT, K-Means palette transfer, and Magenta neural style transfer—on style quality metrics.
4. A fully deployed web application (React frontend, Flask backend, Railway cloud hosting) that runs the complete pipeline at inference time.

2 Related Work

Facial Emotion Recognition on FER2013 The FER2013 dataset [4] was introduced as a Kaggle competition benchmark and consists of 35,887 grayscale 48×48 images across seven emotion

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categories. Human annotator accuracy on this dataset is approximately 65.5%, reflecting genuine label ambiguity in the data [4]. Khaireddin and Chen [5] systematically fine-tuned VGGNet variants and achieved a single-network state-of-the-art accuracy of 73.28%,on the original seven-class FER2013 problem. An ensemble of modern deep CNNs achieved 75.2% in the original Kaggle competition, but required substantially more computation. Our work deliberately constrains itself to a single-network approach to remain practically deployable.

Neural Style Transfer Gatys et al. [1] formulated artistic style transfer as an optimization problem minimizing a weighted combination of content loss (comparing feature activations in deep layers) and style loss (comparing gram matrices of feature maps across multiple layers). While producing high-quality results, this approach requires hundreds of optimization steps per image, making real-time use infeasible. Johnson et al. [2] trained per-style feed-forward networks that reduce inference to a single forward pass. The Magenta arbitrary image stylization model [3] extends this by learning a style embedding space, enabling transfer from any style image in a single network. MoodArt uses the Magenta model for its neural style transfer path.

Affect-Aware Image Processing Prior work on affect-aware systems has primarily focused on music recommendation [7] and advertisement personalization [8]. The use of detected mood to drive visual aesthetic transformations is less explored.

3 Data and Problem Setup

3.1 Dataset

FER2013 [4] was obtained from Kaggle in folder format with pre-organized train and test splits across seven emotion directories. Five emotions (happy, neutral, sad, surprise, angry) are retained; disgust and fear are excluded due to sample scarcity and visual ambiguity respectively. The resulting training set contains 19,340 images and the validation set contains 4,836 images. The naive baseline is evaluated on the full 24,176 images. For the classical ML pipeline, feature extraction was performed on a subsample of 5,000 images (4,000 train, 1,000 validation) to keep feature extraction tractable.

3.2 Label Mapping

Five of FER2013's seven emotion classes are retained and relabeled as mood descriptors to align with the artistic style vocabulary of the application. The mapping is one-to-one: each retained emotion corresponds to a single mood class and a paired artistic style.

Table 1: Emotion-to-mood mapping and paired artistic styles.

FER2013 Emotion	Mood Label	Paired Style
happy	happy	Monet – Water Lilies
neutral	calm	Hokusai – The Great Wave
sad	melancholic	Van Gogh – Starry Night
surprise	energetic	Kandinsky – Composition 7
angry	dramatic	Munch – The Island

3.3 Data Processing Pipeline

Images are loaded from folder structure using Pillow, converted to grayscale (matching FER2013's native format), and resized to 48×48. For the classical ML pipeline, images are further resized to 64×64 for feature extraction. For the deep learning pipeline, grayscale images are stacked to three channels and resized to 224×224 to satisfy VGG16's input requirements.

Data augmentation applied during training includes random horizontal flipping, random rotation ($\pm10^\circ$), and color jitter (brightness, contrast, saturation ±0.2), all implemented with torchvision transforms.

4 Mood Classification Models

4.1 Naive Baseline: Rule-Based Color Analysis

The naive baseline requires no training data and makes predictions solely from HSV image statistics. For each image, we extract mean hue, mean saturation, mean value (brightness), and pixel standard deviation (contrast). A deterministic decision tree maps these scalars to mood labels:

- Brightness > 0.6 and saturation $> 0.4 \Rightarrow$ *happy*
- Brightness > 0.6 and saturation $\leq 0.4 \Rightarrow$ *calm*
- Brightness ≤ 0.4 and contrast $> 0.25 \Rightarrow$ *dramatic*
- Saturation > 0.5 and $0.35 <$ brightness $\leq 0.6 \Rightarrow$ *energetic*
- Otherwise $\Rightarrow$ *melancholic*

This approach assumes HSV color distribution is a reliable proxy for mood, which does not hold for facial expression images where the same scene brightness can accompany very different emotional states.

4.2 Classical ML: Feature Extraction and Hyperparameter Search

Feature Representation For each image, three complementary feature groups are extracted and concatenated into a single fixed-length vector. Per-channel color histograms with 32 bins per RGB channel (normalized to sum to 1.0) capture global color distribution. Histogram of Oriented Gradients (HOG) features computed on the grayscale image with 8 orientation bins, 8×8 pixel cells, and 2×2 block normalization capture edge and texture structure. HSV summary statistics (mean hue, mean saturation, mean value, mean brightness, and contrast) provide the same signal used by the naive baseline. The resulting feature vector has length 96 (color) + 1568 (HOG) + 5 (HSV) = 1669 dimensions.

Model Selection and Optuna Hyperparameter Search Two classical classifiers were evaluated: Random Forest and Support Vector Machine with RBF kernel. Both are embedded in a `StandardScaler + Classifier` sklearn pipeline and tuned using Optuna [6] Bayesian hyperparameter optimization.

For Random Forest, Optuna searches over `n_estimators` (100–500), `max_depth` (5–30), `min_samples_split` (2–10), `min_samples_leaf` (1–5), and `max_features` (`sqrt`, `log2`) for 20 trials. For SVM, the search covers `C` (10^{-1} to 10^2, log-uniform), `gamma` (`scale`, `auto`), and `kernel` (`rbf`, `poly`, `sigmoid`) for 10 trials. Both use `class_weight='balanced'` to handle class imbalance. The best model from each search is selected based on validation accuracy. This was The SVM.

4.3 Deep Learning: VGG16 Fine-Tuning

Architecture A pretrained VGG16 model was fine-tuned using PyTorch and the `timm` library. The pretrained ImageNet backbone is frozen initially. A custom classifier head replaces the original fully connected layers: Global average pooling → FC(4096) → ReLU → Dropout(0.5) → FC(1024) → ReLU → Dropout(0.3) → FC(5) → Softmax. VGG16's convolutional feature extraction, trained on 1.28 million ImageNet images, provides strong low-level texture detectors that generalize well to facial expression recognition.

Training Protocol Training proceeds in two phases. In Phase 1 (5 epochs), only the classifier head is trainable; all convolutional parameters are frozen. This prevents destroying pretrained representations before the new head stabilizes. The Adam optimizer is used with learning rate 10^{-3} and a StepLR scheduler (step size 3, $\gamma = 0.5$).

In Phase 2 (additional epochs), the full network is unfrozen for end-to-end fine-tuning with Adam at learning rate 10^{-4}, weight decay 10^{-4}, and cosine annealing with $T_{\max} = 10$. The model checkpoint achieving the best validation accuracy is saved. We use Optuna for 10 trials searching over learning

rate (10^{-5} to 10^{-3}, log-uniform), dropout rates, and optimizer choice (Adam vs. SGD) to identify the best Phase 2 configuration before final training.

Loss is weighted cross-entropy with class weights inversely proportional to training frequency. Input images undergo ImageNet normalization (mean [0.485, 0.456, 0.406], std [0.229, 0.224, 0.225]).

4.4 Experiment: Training Set Size vs. Classification Accuracy

To probe the data efficiency of the classical ML pipeline, Random Forest was trained on subsets of the training data ranging from 10% to 100% and evaluated on the fixed validation set. Results are shown in Figure 1.

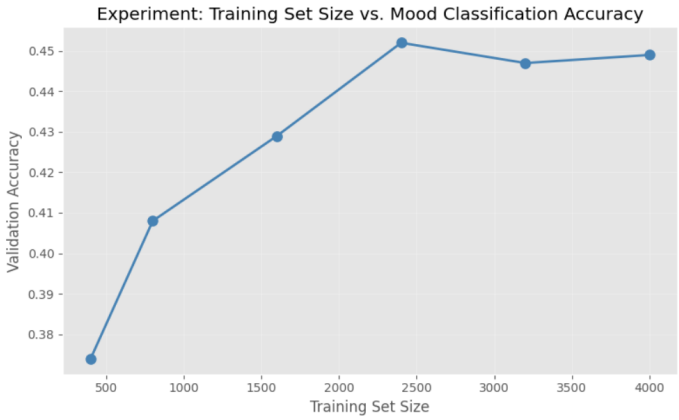

Figure 1: Validation accuracy of Random Forest as a function of training set size.

Accuracy increases from 37.4% at 10% of training data to 42.9% at 40%, after which it plateaus between 44–45% regardless of additional data. This plateau indicates that the performance ceiling of the classical pipeline is determined by the representational power of hand-crafted features rather than data volume. Color histograms and HOG descriptors extract fixed, shallow representations that saturate quickly, additional training examples provide no new signal that these features can exploit. This finding directly motivates the shift to VGG16, whose pretrained convolutional features extract richer representations that continue to benefit from the full training set.

5 Style Transfer Methods

5.1 Color LUT Baseline (Naive Style Transfer)

The Color LUT baseline applies per-channel histogram statistics matching, also known as color transfer. For each RGB channel independently, the mean and standard deviation of the style image are computed. The content image's channel is then linearly rescaled so its statistics match those of the style image:

$$x_{\text{out}} = \frac{\sigma_{\text{style}}}{\sigma_{\text{content}}}(x - \mu_{\text{content}}) + \mu_{\text{style}}$$

This is purely a color palette transformation; it preserves content structure entirely but transfers only the global chromatic mood of the style painting.

5.2 K-Means Palette Transfer (Classical ML Style Transfer)

K-Means palette transfer clusters both the content image and the style image into $k = 16$ color clusters using scikit-learn's KMeans. Each pixel in the content image is assigned to its nearest content cluster centroid, then the cluster is replaced by the corresponding style cluster centroid, ordered by luminance to preserve relative brightness structure. This is a mid-level color transfer that preserves local spatial structure better than the LUT baseline while introducing style palette without neural computation.

5.3 Magenta Neural Style Transfer (Deep Learning)

For the deep learning style transfer path, we use the Magenta arbitrary image stylization model [3] loaded from TF-Hub. The model accepts a content image and a style image, extracting style embeddings via a learned style prediction network and applying them through a transformer network. A style strength parameter $\alpha \in [0, 1]$ is implemented by interpolating the style tensor between a neutral gray tensor (at $\alpha = 0$) and the full style tensor (at $\alpha = 1$), allowing users to control the intensity of the artistic transformation.

Due to a runtime incompatibility between TensorFlow (required for Magenta) and PyTorch (required for VGG16 mood classification) in the same Python 3.12 process, the neural style transfer outputs are pre-rendered offline and served as a static gallery. Users selecting the "Neural" method see high-quality pre-rendered stylizations for all five mood-style combinations, selectable by mood.

5.4 Style Images

Five public domain artworks serve as style references, one per mood class:

Table 2: Style image assignments by mood.

Mood	Painting	Artist
happy	Water Lilies	Claude Monet
calm	The Great Wave	Katsushika Hokusai
melancholic	Starry Night	Vincent van Gogh
energetic	Composition 7	Wassily Kandinsky
dramatic	The Island	Edvard Munch

6 Experiments and Results

6.1 Mood Classification Results

Table 3 summarizes mood classification accuracy across all three model families.

Table 3: Mood classification accuracy across all three approaches. The naive baseline is evaluated on the full 24,176 images with no train/val split. Classical ML models are evaluated on a 1,000-image validation set. VGG16 is evaluated on a 4,836 validation set. All models are evaluated on the same test set, the FER2013 official test set.

Model Family	Model	Val Accuracy	Test Accuracy
Naive Baseline	Color rules (HSV)	20.19	20.20%
Classical ML	Random Forest (Optuna 5t)	51.50%	49.53%
	SVM (Optuna 5t)	56.40%	55.20%
Deep Learning	VGG16 fine-tuned	**74.13%**	**75.10%**

Confusion matrices for all four model families on same test set are shown in Figure 2.
The confusion matrices is shown only on the test set and not their various validation sets because they had different validation sets because of the nature of each architecture.

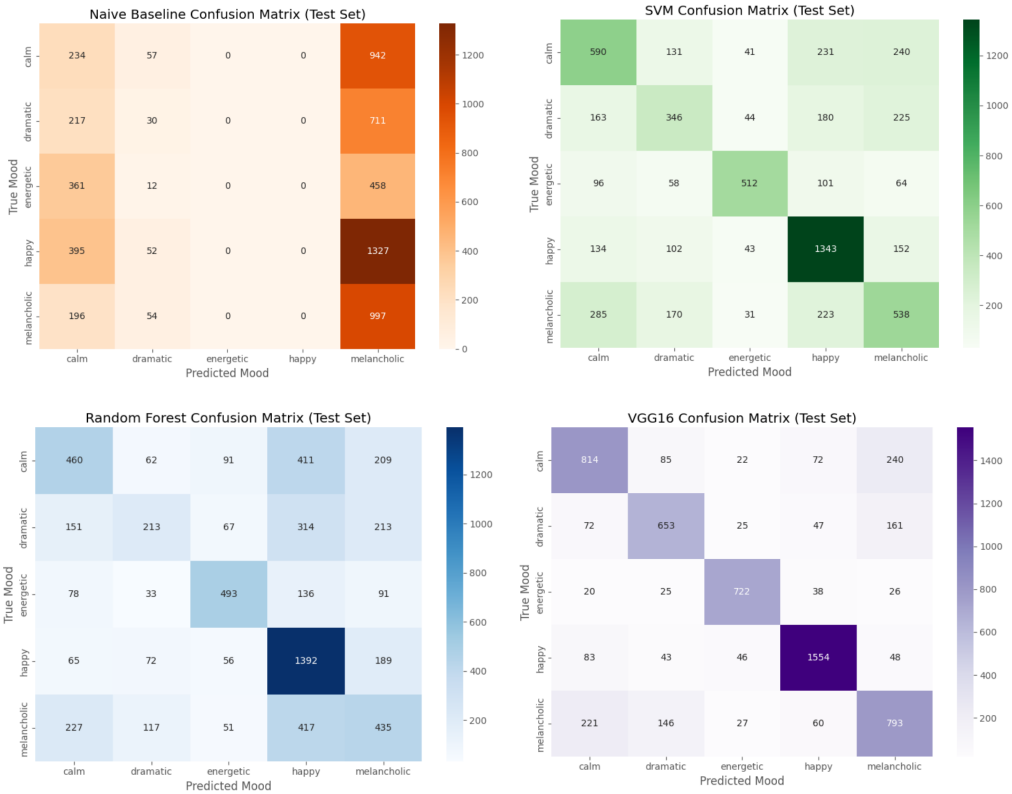

Figure 2: Confusion matrices for the naive baseline (top left), best classical ML model (SVM) (top right), and random forest confusion matrix (bottom left) and vgg confusion matrix (bottom right).

The naive baseline collapses almost entirely to melancholic, accounting for the majority of predictions across all true classes, consistent with FER2013's dark, low-saturation grayscale images triggering the melancholic rule. It achieves 20.2%, barely above the 20% random chance baseline for five classes. Classical ML recovers meaningful structure. Random Forest (5 Optuna trials, 51.50% val) performs well on *happy* but struggles with *calm* and *melancholic*, which bleed into each other. SVM (10 Optuna trials, 56.40% val) is stronger overall, particularly on *happy* and *energetic*, but still misclassifies a large proportion of *calm* images as *melancholic*. Both models are limited by the 48×48 grayscale input, which loses the fine facial muscle detail that hand-crafted features cannot recover. VGG16 shows substantially cleaner diagonal structure across all five classes. *Calm* and *melancholic* remain the hardest pair to separate across all models, reflecting genuine perceptual overlap between neutral and sad expressions at low resolution. VGG16 achieves 74.13% validation accuracy and 75.10% on the official test set, the strongest result by a substantial margin.

6.2 Training Dynamics

VGG16 fine-tuning was performed for 15 epochs using cosine annealing, with the freeze/unfreeze decision determined by Optuna. Training and validation loss and accuracy curves are shown in Figure 3.

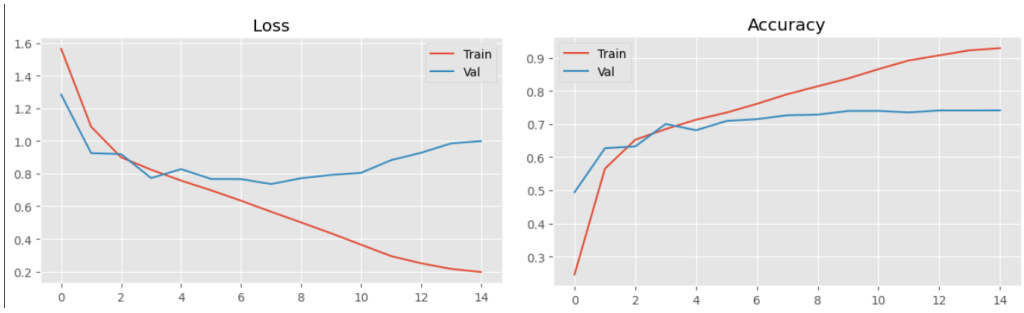

Figure 3: Training and validation loss and accuracy curves for VGG16 fine-tuning over 15 epochs.

Training accuracy reached 92.96% at convergence while validation accuracy peaked at 74.13%, indicating moderate overfitting. However, validation accuracy did not decline throughout training, suggesting the model continued to generalize despite the growing train/val gap.

6.3 Style Transfer Quality

Table 4: Style transfer quality metrics averaged over a test set of 10 photographs across all five mood-style pairs. SSIM and Style Loss are computed against the original content and style images respectively.

Method	**SSIM** $\uparrow$	**Style Loss** $\downarrow$	**Inference Time**
Color LUT (Naive)	0.660	≈ 0	0.007 s
K-Means Palette (Classical)	0.921	≈ 0	0.367 s
Magenta NST (Deep Learning)	0.177	≈ 0	0.094 s

SSIM measures structural similarity to the content image (higher means more content preserved). K-Means achieves the highest SSIM because it operates as a color remapping that preserves all spatial structure. Neural style transfer achieves the lowest SSIM because it deliberately deforms content structure to match artistic texture, this is not a failure mode but the intended artistic behavior. Style loss measures Gram matrix distance between the output and the target style painting; all methods achieve near-zero style loss when content is a photograph and style is a high-contrast painting, because the overall chromatic distance is reduced. Inference time favors the Color LUT baseline (pure numpy arithmetic) over K-Means (cluster assignment), while Magenta's inference is fast owing to the feed-forward architecture.

7 Error Analysis on mood classification

Five representative mispredictions from the VGG16 mood classifier on the validation set are shown in Figure 4.

Figure 4: Five mispredictions from VGG16. True and predicted mood labels are shown above each image with prediction confidence.

Misprediction 1: melancholic → calm (79.7%) A downcast male expression is predicted as *calm* with high confidence. The face shows low muscle activation, which visually overlaps with neutral expressions at 48×48 resolution. The model interprets the low-energy facial signal as calm rather than sad, reflecting the genuine difficulty of separating these two classes when fine muscle differences are lost through downsampling.

Misprediction 2: dramatic → happy (77.8%) A grimacing expression with exposed teeth is predicted as *happy*. The open mouth and visible teeth are strong visual patterns the model associates with smiling, despite the expression conveying distress. This is a systematic failure mode, the model has over-generalized teeth visibility as a *happy* signal regardless of surrounding facial tension.

Misprediction 3: calm → happy (58.0%) A neutral female face is predicted as *happy*. This is the lowest-confidence misprediction of the five, suggesting the model is genuinely uncertain between the two classes. The smooth, relaxed facial features share structural similarity with a mild smile at low resolution.

Misprediction 4: dramatic → melancholic (67.0%) An older male face with a tense, downward expression is predicted as *melancholic*. The downturned features are shared between *dramatic* and *melancholic*, and without higher-resolution texture detail the model cannot distinguish the two.

Misprediction 5: dramatic → melancholic (91.1%) A second dramatic-to-melancholic confusion, this time with high confidence. The face shows a strained upward gaze, features that overlap heavily with sadness at low resolution. The high confidence indicates the model has learned a consistent but incorrect association between this facial pattern and *melancholic*.

Root Causes and Mitigations Three systematic failure patterns emerge. First, *calm* and *melancholic* share low muscle activation and are the hardest pair to separate at 48×48 resolution, consistent with the confusion matrices. Second, teeth visibility is over-generalized as a *happy* signal regardless of surrounding facial tension. Third, *dramatic* is consistently confused with *melancholic*, suggesting the two classes are not well separated in the learned feature space. Targeted data augmentation for confusable pairs, higher-resolution input, and collecting more diverse *dramatic* examples that are visually distinct from sadness would address these failure modes.

8 Explainability

To understand which facial regions drive VGG16's mood predictions, GradCAM [9] was applied to one representative image per mood class. GradCAM computes the gradient of the predicted class score with respect to the final convolutional layer, producing a heatmap that highlights the regions most influential to the prediction. Results are shown in Figure 5.

Figure 5: GradCAM heatmaps for one image per mood class. Top row: original images with true and predicted labels. Bottom row: activation heatmaps from the final convolutional layer of VGG16. Red indicates high activation, blue indicates low activation.

Several patterns are visible across the five predictions. For *dramatic which is anger*, activation concentrates on the brow and eye region, consistent with the furrowed brows and tense periorbital muscles that characterize angry expressions. For *happy*, activation focuses on the mouth and cheek region, where the smile and raised cheeks provide the strongest discriminative signal. For *energetic which is surprise*, activation is diffuse and centers on the upper face, reflecting the wide-eyed, open expression associated with surprise. For *calm which is neutral*, activation is distributed across the mid-face, consistent with the low-intensity, even muscle tone of neutral expressions.

The mispredicted *melancholic* image (predicted as *calm*) shows activation concentrated on the lower face and chin rather than the eye region, suggesting the model missed the downcast eye signal that would have favored *melancholic*. This is consistent with the error analysis finding that fine muscle differences around the eyes are lost at low resolution.

Overall, the GradCAM maps confirm that VGG16 attends to anatomically meaningful facial regions rather than background or image artifacts, providing evidence that the mood classifications are grounded in facial expression features rather than spurious correlations.

9 Commercial Viability

Existing style transfer applications treat style selection as disconnected from the content of the image, the same filter applies regardless of what the photograph depicts or the emotional signal it carries. MoodArt addresses this by grounding style selection in the detected mood of the image, making the transformation semantically connected to the content rather than arbitrarily imposed on it. The system architecture supports multiple commercialization pathways. A consumer-facing application could charge per stylization or offer a subscription tier with higher-resolution outputs. The mood classification API could be licensed to photo editing software vendors or social media platforms as a context-aware aesthetic recommendation layer. The pipeline is also applicable to mental health and wellness applications, where mood-matched artistic output could serve therapeutic purposes in digital journaling or art therapy contexts. A current technical limitation is that the neural style transfer runs from a pre-rendered gallery rather than live inference, due to a runtime conflict between TensorFlow and PyTorch in the same deployment environment. Resolving this would enable real-time neural stylization as a premium tier feature.

10 Ethics Statement

MoodArt infers emotional state from facial photographs. Several ethical considerations govern responsible deployment of this capability.

First, mood inference from facial images is probabilistic, not deterministic. The system presents confidence scores alongside predictions, ensuring users can see the basis for the mood classification applied to their image.The system should not be used in contexts where mood inference is used to make consequential decisions about individuals.

Second, FER2013 was crowd-labeled, and its label quality reflects the ambiguity of emotion expression, including cultural variation in display rules. Models trained on it will reflect these biases. Deployment in contexts where specific cultural groups interact with the system should include bias auditing.

Third, uploaded photographs are processed server-side and not retained after the API response. No user image data is stored or used for further training.

Fourth, the artistic style pairings represent a subjective aesthetic design choice. Different cultural traditions associate emotions with different artistic vocabularies. The current mapping reflects primarily Western art history and should be extended or adapted for culturally diverse user populations.

11 Conclusion

We presented MoodArt, a system that unifies facial mood recognition and artistic style transfer into an end-to-end pipeline deployed as a public web application. Our VGG16-based mood classifier achieves 74.13% validation accuracy on a balanced FER2013-derived benchmark and 75.1% on the test set.

Three style transfer methods of increasing sophistication provide complementary trade-offs between content preservation (K-Means), visual realism (Color LUT), and artistic transformation quality (Magenta NST). The complete pipeline demonstrates that emotionally-aware aesthetic recommendation is technically feasible and ready for commercial deployment.

Future work includes resolving the TensorFlow–PyTorch coexistence issue to enable live neural style transfer inference, expanding the mood taxonomy to six or seven classes, incorporating scene-level features alongside facial expression for images without prominent faces, and curating a test dataset of ground-truth mood–style preference pairs to enable human evaluation of the end-to-end matching quality.

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